

Chapter 9

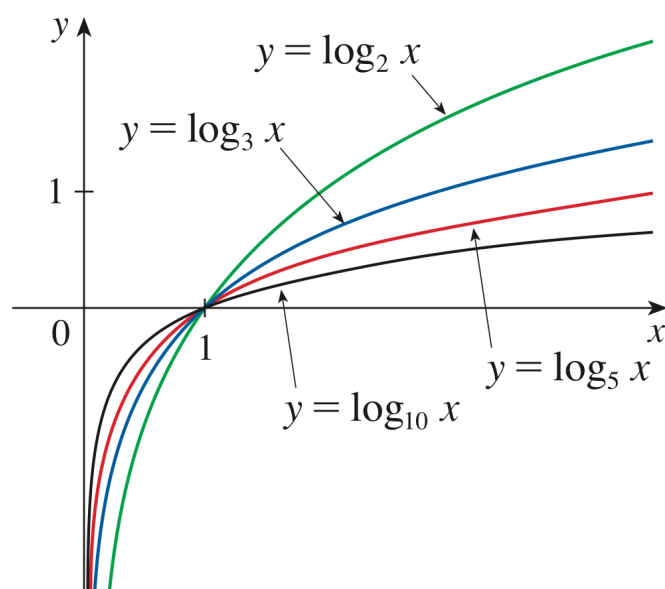
Derivatives of Logarithmic and Inverse Trigonometric Functions

§ Section: 1.5,3.6

9.1 Logarithmic Functions

If $a > 0$ and $a \neq 1$, the exponential function $f(x) = a^x$ is either increasing or decreasing and so it is **one-to-one** by the Horizontal Line Test. It therefore has an inverse function which is called the **logarithmic function with base a** and is denoted by $\log_a x$.

The logarithm with base e is called the **natural logarithm** and has a special notation: $\log_e x = \ln x$.



Change of Base Formula

For any positive number $a (a \neq 1)$, we have

$$\log_a x = \frac{\ln x}{\ln a}$$

Example 1.1

Compute $\frac{d}{dx} \ln x = \frac{1}{x}$

Sol: Let $y = \ln x$. Then $e^y = x$.

$$e^{y(x)} = x \quad \xRightarrow{\frac{d}{dx}} \quad e^{y(x)} \cdot y'(x) = 1 \quad \Rightarrow \quad y'(x) = \frac{1}{e^{y(x)}} = \frac{1}{x}.$$

Example 1.2

Compute $\frac{d}{dx} \log_a x = \frac{d}{dx} \left(\frac{\ln x}{\ln a} \right) = \frac{1}{\ln a} \cdot \frac{1}{x}$

$$\frac{d}{dx} (a^x) = \ln a \cdot a^x$$

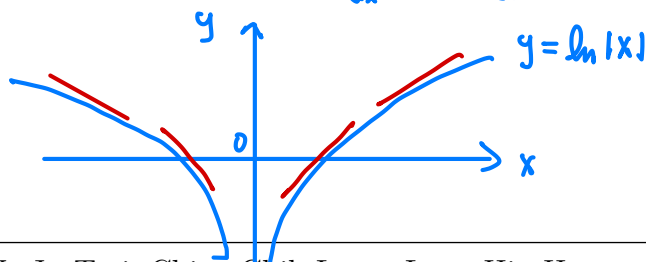
Example 1.3

Compute $\frac{d}{dx} \ln |x| = \frac{1}{x}$

Sol:

If $x > 0$, then $\ln |x| = \ln x$, $\frac{d}{dx} \ln |x| = \frac{d}{dx} \ln x = \frac{1}{x}$.

If $x < 0$, then $\ln |x| = \ln(-x)$, $\frac{d}{dx} \ln |x| = \frac{d}{dx} \ln(-x) = \frac{-1}{-x} = \frac{1}{x}$



Example 1.4

Compute $\frac{d}{dx} \ln |f(x)| = \frac{d}{dx} \ln |f(x)|$

$$= \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}$$

Derivatives of Logarithmic Functions

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} \log_a x = \frac{1}{x \ln a}, \quad \frac{d}{dx} \ln |x| = \frac{1}{x}$$

Combine the above formulas with the chain rule

$$\frac{d}{dx} \ln g(x) = \frac{g'(x)}{g(x)}, \quad \frac{d}{dx} \ln |g(x)| = \frac{g'(x)}{g(x)}$$

9.2 Logarithmic Differentiation

The calculation of derivatives of complicated functions involving **products, quotients, or powers** can often be simplified by **taking logarithms**. This method is called *logarithmic differentiation*.

Steps in Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the properties of logarithms to simplify.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for y' .

Example 2.1

Prove the power rule $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$ for any $n \in \mathbb{R}$.

Sol: Let $f(x) = x^n$. $\ln|f(x)| = \ln(|x|^n) = n \cdot \ln|x|$

$$\Rightarrow \frac{\frac{d}{dx} f(x)}{f(x)} = n \cdot \frac{1}{x}$$

$$\Rightarrow f'(x) = n \cdot \frac{1}{x} \cdot f(x) = n \cdot \frac{1}{x} \cdot x^n = n \cdot x^{n-1}.$$

Example 2.2

Let $f(x) = x^x$, for $x > 0$. Find $f'(x)$.

$f(x) = x^x \cdot \ln x$ X
 $f'(x) = x \cdot x^{x-1} = x^x$ X

Sol 1:

$$\ln f(x) = \ln(x^x) = x \cdot \ln x.$$

$$\Rightarrow \frac{\frac{d}{dx} f(x)}{f(x)} = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1.$$

Sol 2: $f(x) = x^x = e^{\ln x \cdot x}$

$$\Rightarrow f'(x) = x^x (\ln x + 1)$$

Example 2.3

Let $f(x) = x^{2^x} + x^{x^2} + 2^{x^x}$, for $x > 0$. Find $f'(x)$.

Sol: ① Let $g(x) = x^{2^x}$. $\ln g(x) = \ln(x^{2^x}) = 2^x \cdot \ln x$

$$\Rightarrow \frac{\frac{d}{dx} g(x)}{g(x)} = 2^x \cdot \ln 2 \cdot \ln x + 2^x \cdot \frac{1}{x}$$

$$\Rightarrow g'(x) = x^{2^x} (\ln 2 \cdot 2^x \cdot \ln x + 2^x \cdot \frac{1}{x})$$

$$\textcircled{3} \frac{d}{dx} (2^{x^x}) = \ln 2 \cdot 2^{x^x} \cdot (x^x)' = \ln 2 \cdot 2^{x^x} \cdot x^x (\ln x + 1).$$

② Let $h(x) = x^{x^2}$.

$$\ln h(x) = \ln(x^{x^2}) = x^2 \cdot \ln x.$$

$$\Rightarrow \frac{\frac{d}{dx} h(x)}{h(x)} = 2x \cdot \ln x + x^2 \cdot \frac{1}{x} = 2x \cdot \ln x + x$$

$$h'(x) = x^{x^2} (2x \cdot \ln x + x)$$

Example 2.4

Derive $\frac{d}{dx}[f(x)^{g(x)}] = f(x)^{g(x)} \left[g'(x) \cdot \ln|f(x)| + g(x) \cdot \frac{f'(x)}{f(x)} \right]$.

Let $h(x) = f(x)^{g(x)}$.

$$\ln|h(x)| = \ln(|f(x)|^{g(x)}) = g(x) \cdot \ln|f(x)|.$$

$$\Rightarrow \frac{\frac{d}{dx} h(x)}{h(x)} = g'(x) \cdot \ln|f(x)| + g(x) \cdot \frac{f'(x)}{f(x)}$$

Example 2.5

Compute $\frac{d}{dx} [(1+x^3)^x (\ln x)^{\cos x}]$, for $x > 1$.

Sol: Let $f(x) = (1+x^3)^x \cdot (\ln x)^{\cos x}$

$$\ln f(x) = x \ln(1+x^3) + \cos x \cdot \ln(\ln x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = 1 \cdot \ln(1+x^3) + x \cdot \frac{3 \cdot x^2}{1+x^3} + \cos x \cdot \frac{1}{\ln x} \cdot \frac{1}{x} - \sin x \cdot \ln(\ln x).$$

$$\text{Hence } f'(x) = f(x) \cdot \left[\ln(1+x^3) + \frac{3x^3}{1+x^3} + \frac{\cos x}{x \cdot \ln x} - \sin x \cdot \ln(\ln x) \right]$$

Example 2.6

Find $\frac{d}{dx} (\log_{f(x)} g(x)) = \frac{d}{dx} \left(\frac{\ln g(x)}{\ln f(x)} \right)$

$$= \frac{\frac{g'(x)}{g(x)} \cdot \ln f(x) - \ln g(x) \cdot \frac{f'(x)}{f(x)}}{(\ln f(x))^2}$$

$$\text{Ex: } \frac{d}{dx} \left(\frac{(x^2-3x+1)^3 e^{2x}}{\sqrt{x^2+x+1} \cdot (x^3-1)^2} \right)$$

" f(x)

$$\ln f(x) = 3 \ln |x^2-3x+1| + 2x - \frac{1}{2} \ln(x^2+x+1) - 2 \ln |x^3-1|$$

$$\frac{d}{dx} f(x)^{g(x)} = ? \quad \frac{d}{dx} \log_{f(x)} g(x) = ?$$

9.3 The number e as a Limit

The number e expressions Known: $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$.

$$e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = \lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x$$

|| let $y = \frac{1}{x}$

$$\lim_{y \rightarrow 0^+} (1+y)^{\frac{1}{y}} = e$$

Example 3.1

Try to compute $\left. \frac{d}{dx} \ln x \right|_{x=1} = \left. \frac{1}{x} \right|_{x=1} = 1$.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\ln(1+h) - \ln 1}{h} &= \lim_{h \rightarrow 0} \frac{1}{h} \ln(1+h) \\ &= \lim_{h \rightarrow 0} \ln(1+h)^{\frac{1}{h}} = \ln \left(\lim_{h \rightarrow 0} (1+h)^{\frac{1}{h}} \right) = 1 \\ &\Rightarrow \lim_{h \rightarrow 0} (1+h)^{\frac{1}{h}} = e. \end{aligned}$$

Example 3.2

Compute $\lim_{n \rightarrow \infty} \left(1 - \frac{5}{n}\right)^{3n}$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\frac{-n}{5}}\right)^{\frac{-n}{5} \cdot (-15)} \\ \stackrel{=}{=} \lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^{x \cdot (-15)} = e^{-15} \end{aligned}$$

Example 3.3

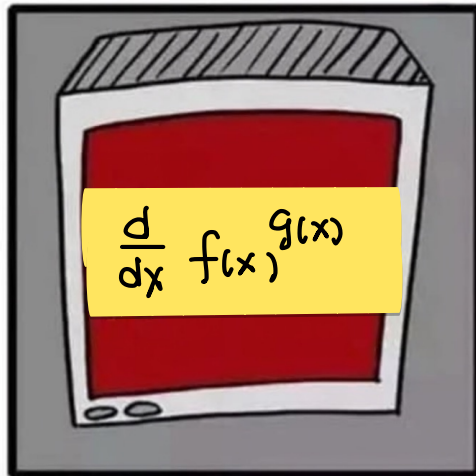
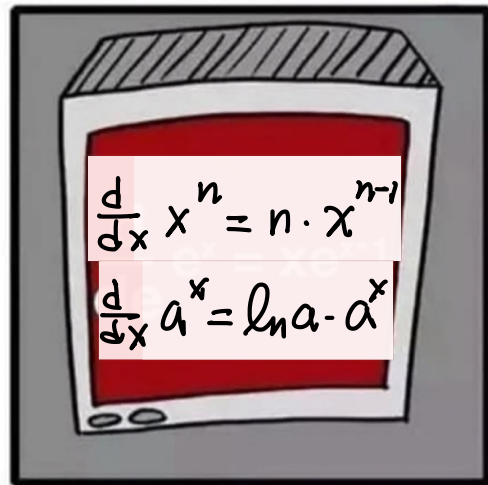
Find $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$.

For $x \neq 0$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\frac{n}{x}}\right)^{\frac{n}{x} \cdot x} \\ \stackrel{=}{=} \lim_{y \rightarrow \pm\infty} \left(1 + \frac{1}{y}\right)^{y \cdot x} = e^x. \end{aligned}$$

For $x = 0$,

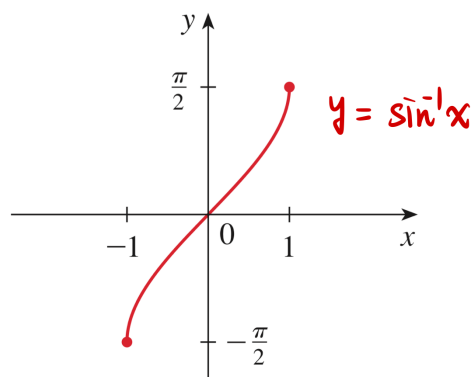
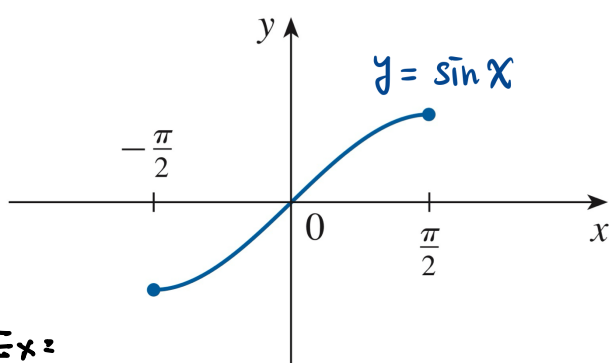
$$\lim_{n \rightarrow \infty} \left(1 + \frac{0}{n}\right)^n = \lim_{n \rightarrow \infty} 1^n = 1 = e^0$$



9.4 Inverse Trigonometric Functions

Define $\arcsin x$, $\sin^{-1} x$:

$$\sin^{-1} x = y \iff \textcircled{1} \sin y = x \text{ and } \textcircled{2} -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



Ex:

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

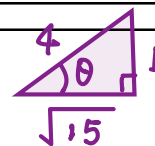
$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

$$\sin^{-1}(\sin \pi) = 0$$

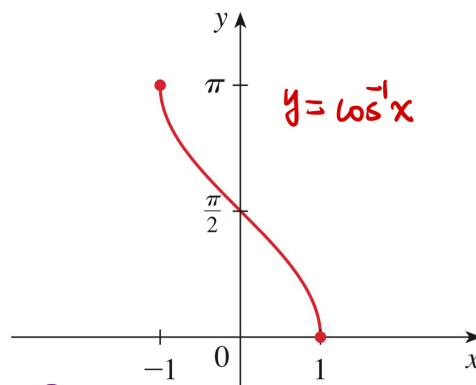
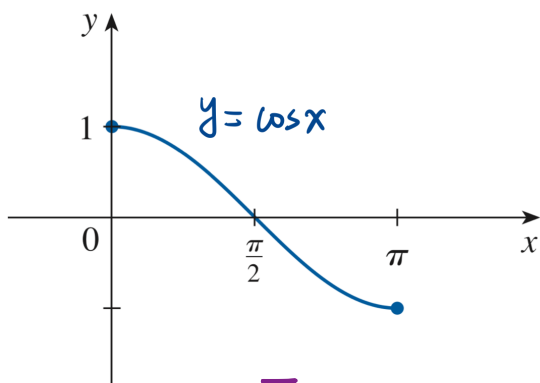
$$\tan(\sin^{-1} \frac{1}{4}) = \frac{1}{\sqrt{15}}$$

Define $\arccos x$, $\cos^{-1} x$:

Let $\theta = \sin^{-1}(\frac{1}{4})$
 $\sin \theta = \frac{1}{4}$



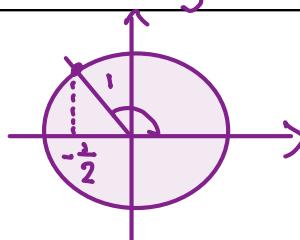
$$\cos^{-1} x = y \iff \cos y = x \text{ and } 0 \leq y \leq \pi$$



Ex: $\cos^{-1}(0) = \frac{\pi}{2}$

$$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2}{3}\pi$$

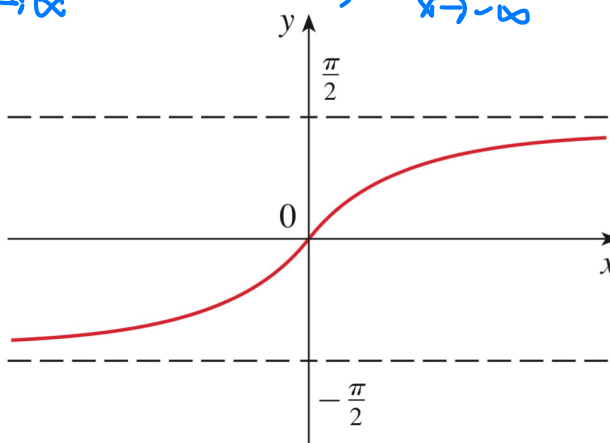
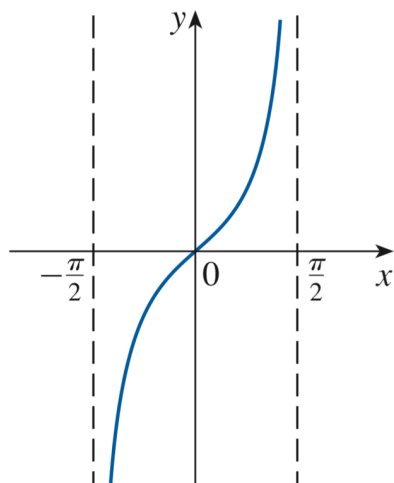
$$\sec(\cos^{-1} \frac{1}{3}) = 3$$



Define $\arctan x$, $\tan^{-1} x$:

$$\tan^{-1} x = y \iff \tan y = x \text{ and } -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$$\lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}, \quad \lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}.$$



$$y = \tan x$$

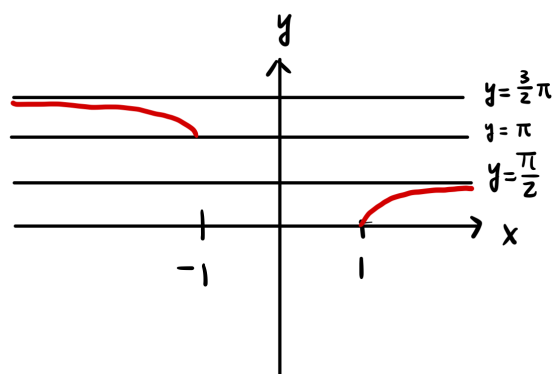
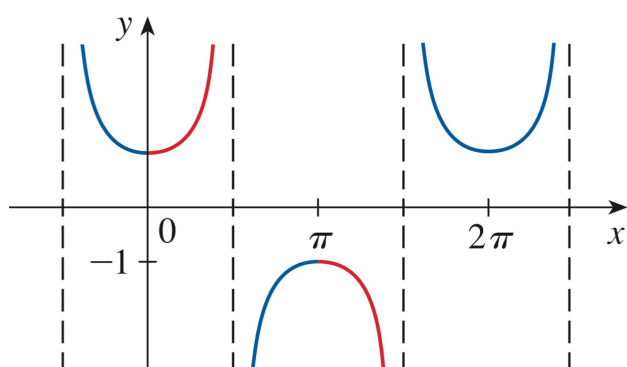
$$y = \tan^{-1} x$$

Ex: $\tan^{-1}(1) = \frac{\pi}{4}$

$\tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$

Define $\sec^{-1} x$:

$$\sec^{-1} x = y \iff \sec y = x \text{ and } y \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$$



$$y = \sec x$$

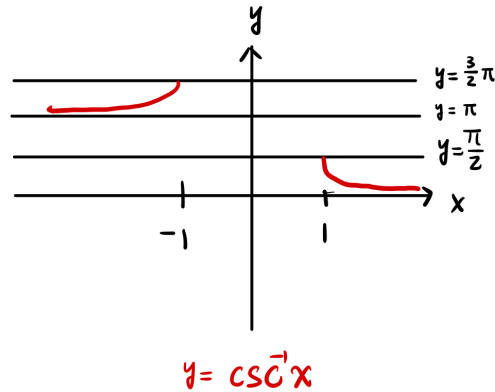
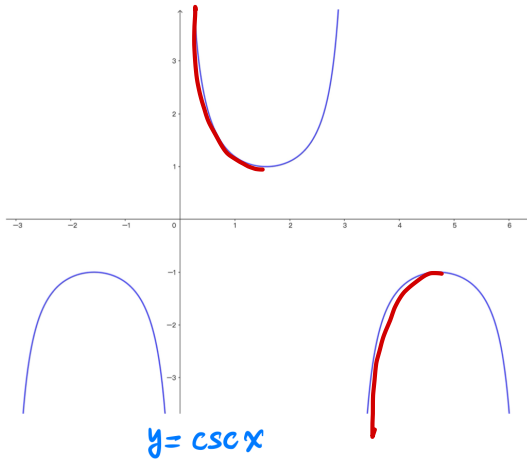
$$y = \sec^{-1} x$$

Ex: $\sec^{-1}(\sqrt{2}) = \frac{\pi}{4}$

$\sec^{-1}(-\sqrt{2}) = \frac{5}{4}\pi$

Define $\csc^{-1} x$:

$$\csc^{-1} x = y \iff \csc y = x \text{ and } y \in (0, \frac{\pi}{2}] \cup (\pi, \frac{3}{2}\pi]$$



9.5 Compositions of trigonometric functions and inverse trigonometric functions

Example 5.1

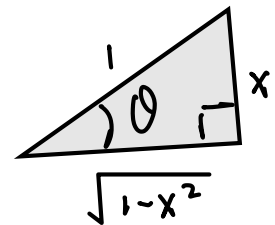
Simplify $\cos(\sin^{-1} x)$. $= \sqrt{1-x^2}$

Sol: Let $\theta = \sin^{-1} x \iff \sin \theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Find $\cos(\theta)$. $\cos^2 \theta = 1 - \sin^2 \theta = 1 - x^2$.

$\Rightarrow \cos \theta = \pm \sqrt{1-x^2}$. $\because \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \therefore \cos \theta \geq 0$

$\Rightarrow \cos \theta = \cos(\sin^{-1} x) = \sqrt{1-x^2}$.



Example 5.2

Simplify $\tan(\sec^{-1} x)$. $= \sqrt{x^2-1}$

Sol: Let $\theta = \sec^{-1} x \iff \sec \theta = x$ and $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3}{2}\pi)$

Find $\tan \theta$. $\tan^2 \theta + 1 = \sec^2 \theta$

$\Rightarrow \tan^2 \theta = \sec^2 \theta - 1 = x^2 - 1$

$\Rightarrow \tan \theta = \pm \sqrt{x^2-1}$. $\because \theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3}{2}\pi)$

Example 5.3Simplify $\tan(\sin^{-1} x)$.Let $\theta = \sin^{-1} x \Leftrightarrow \sin \theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.Find $\tan \theta = \frac{\sin \theta}{\cos \theta}$. Note that $\sin \theta = x$ and $\cos \theta = \sqrt{1-x^2}$. Hence $\tan \theta = \frac{x}{\sqrt{1-x^2}}$.**9.6 Derivatives of Inverse Trigonometric Functions****Example 6.1**Derive $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$ sol: Let $y = \sin^{-1} x \Leftrightarrow \sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

$$\sin y(x) = x \xRightarrow{\frac{d}{dx}} \cos y \cdot y'(x) = 1$$

$$\Rightarrow y'(x) = \frac{1}{\cos y} = \frac{1}{\cos(\sin^{-1} x)} = \frac{1}{\sqrt{1-x^2}}$$

Example 6.2Derive $\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$ sol: Let $y = \cos^{-1} x \Leftrightarrow \cos y = x$ and $y \in [0, \pi]$

$$\cos y(x) = x \xRightarrow{\frac{d}{dx}} -\sin y \cdot y' = 1 \Rightarrow y'(x) = \frac{-1}{\sin y} = \frac{-1}{\sqrt{1-x^2}}$$

$$\because \cos y = x \quad \therefore \sin^2 y = 1-x^2, \quad \sin y = \pm \sqrt{1-x^2}$$

$$\because y \in [0, \pi] \quad \therefore \sin y \geq 0 \Rightarrow \sin y = \sqrt{1-x^2}$$

Example 6.3

Derive $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$

sol: Let $y = \tan^{-1} x \Leftrightarrow \tan y = x$ and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$\tan y(x) = x \xrightarrow{\frac{d}{dx}} \sec^2 y \cdot y' = 1$$

$$\Rightarrow y' = \frac{1}{(\sec y)^2} = \frac{1}{1+x^2}.$$

Example 6.4

Derive $\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$

sol: Let $y = \sec^{-1} x \Leftrightarrow \sec y = x$ and $y \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3}{2}\pi)$

$$\sec y(x) = x \xrightarrow{\frac{d}{dx}} \tan y \cdot \sec y \cdot y' = 1$$

$$\Rightarrow y' = \frac{1}{\sec y \cdot \tan y} = \frac{1}{x\sqrt{x^2-1}}$$

Derivatives of Trigonometric Functions:

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

Example 6.5

Compute $\frac{d}{dx} \tan^{-1}(ax)$ where $a \in \mathbb{R}$, $a \neq 0$.

$$\frac{1}{1+(ax)^2} \cdot a = \frac{a}{1+(ax)^2}.$$

Example 6.6

Compute $\frac{d}{dx}(\sin^{-1}(\sqrt{x}))^2$.

$$\begin{aligned}
 \frac{d}{dx} (\sin^{-1}(\sqrt{x}))^2 &= 2 (\sin^{-1} \sqrt{x}) \cdot (\sin^{-1} \sqrt{x})' \\
 &= 2 \sin^{-1}(\sqrt{x}) \cdot \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} \\
 &= \frac{\sin^{-1}(\sqrt{x})}{\sqrt{x-x^2}}.
 \end{aligned}$$

Example 6.7

$\frac{d}{dx} \tan^{-1}(x - \sqrt{1+x^2})$.

$$\begin{aligned}
 &\frac{1}{1+(x-\sqrt{1+x^2})^2} \cdot \left(1 - \frac{2x}{2\sqrt{1+x^2}}\right) \\
 &= \frac{1}{2(1+x^2) - 2x\sqrt{1+x^2}} \cdot \frac{\sqrt{1+x^2} - x}{\sqrt{1+x^2}} \\
 &= \frac{1}{2} \cdot \frac{1}{\sqrt{1+x^2} (\cancel{\sqrt{1+x^2} - x})} \cdot \frac{\cancel{\sqrt{1+x^2} - x}}{\sqrt{1+x^2}} = \frac{1}{2(1+x^2)}.
 \end{aligned}$$